

Basketball Computer Vision Pipeline: Multi-View Tracking & Fusion

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A multi-camera computer vision pipeline that detects, tracks, and re-identifies basketball players and the ball across four synchronized camera feeds, fusing local track identities into a single set of persistent Global IDs.

Overview

This repository contains the complete codebase and output deliverables for a multi-camera basketball tracking assignment. The pipeline ingests four synchronized camera angles — **Front-Left**, **Near-Left**, **Front-Right**, and **Near-Right** — and produces a unified, cross-camera tracking dataset by:

1. Detecting players and the basketball in each view
 2. Assigning local (per-camera) tracking IDs
 3. Fusing those local identities across all four views using appearance-based re-identification
 4. Outputting a master dataset with persistent **Global IDs**
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Architecture & Model Choices

The pipeline is built as a multi-stage CV system:

Stage	Model / Method	Purpose
Object Detection	YOLOv8s	Detects <code>class 0</code> (person) and <code>class 32</code> (sports ball) across 1080p frames, balancing real-time speed with accuracy
Intra-Camera	ByteTrack / DeepSORT	Performs frame-to-frame tracking within a single camera view, maintaining persistent local track IDs through brief occlusions

Tracking		
Cross-Camera Fusion & Re-ID	ResNet-50 (truncated)	Extracts 2048-dimensional appearance embeddings from detection crops; a cosine similarity matrix (threshold 0.60) matches identities across the four camera feeds and merges them into a single Global ID

Repository Structure

```

basketball-cv-tracking/
├── basketball_cv.ipynb      # End-to-end pipeline: detection, tracking, and Re-ID f
├── output/                  # Generated tracking outputs / deliverables
└── README.md

```

Setup & Execution

The entire pipeline is designed to run in a single Google Colab notebook to take advantage of free GPU acceleration.

1. Clone the repository

```
git clone https://github.com/syedsaliq7866/basketball-cv-tracking.git
```

2. **Open the notebook** Open `basketball_cv.ipynb` in Google Colab and set the hardware accelerator to **T4 GPU** (Runtime → Change runtime type → T4 GPU).

3. **Mount Google Drive** (if your video data is stored there) and install dependencies:

```
pip install ultralytics opencv-python supervision pandas
```

4. **Run all cells** to execute detection, intra-camera tracking, and cross-camera Re-ID fusion in sequence.

Output

The pipeline produces a master tracking dataset containing, per frame and per camera view:

- Bounding boxes for detected players and the ball
- Local track IDs (per camera)
- Fused Global IDs (consistent across all four camera views)

Generated artifacts are saved to the `output/` directory.

Requirements

- Python 3.8+
 - GPU (T4 or better recommended)
 - `ultralytics`
 - `opencv-python`
 - `supervision`
 - `pandas`
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License

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